

RoboSim

A 3D Robot Navigation Simulator

By David Wusu, Ashby Thorpe, Daniel Zhang and Asher Bloom

Our Project

- Client: Marius Jurt – Engineer at Toshiba
- Business Problem: Explore a new type of robotic box that could revolutionise warehouse management with self-moving and self-stacking capabilities
- Objective: Create (improve) a platform to do the following:
 - Allow 3D routing algorithms to be benchmarked
 - Enable experimentation of box configurations



User Stories

Warehouse Managers

As a **Warehouse Manager**, I want to **optimise** storage and retrieval of goods to **minimise** management overheads

Algorithm Designers

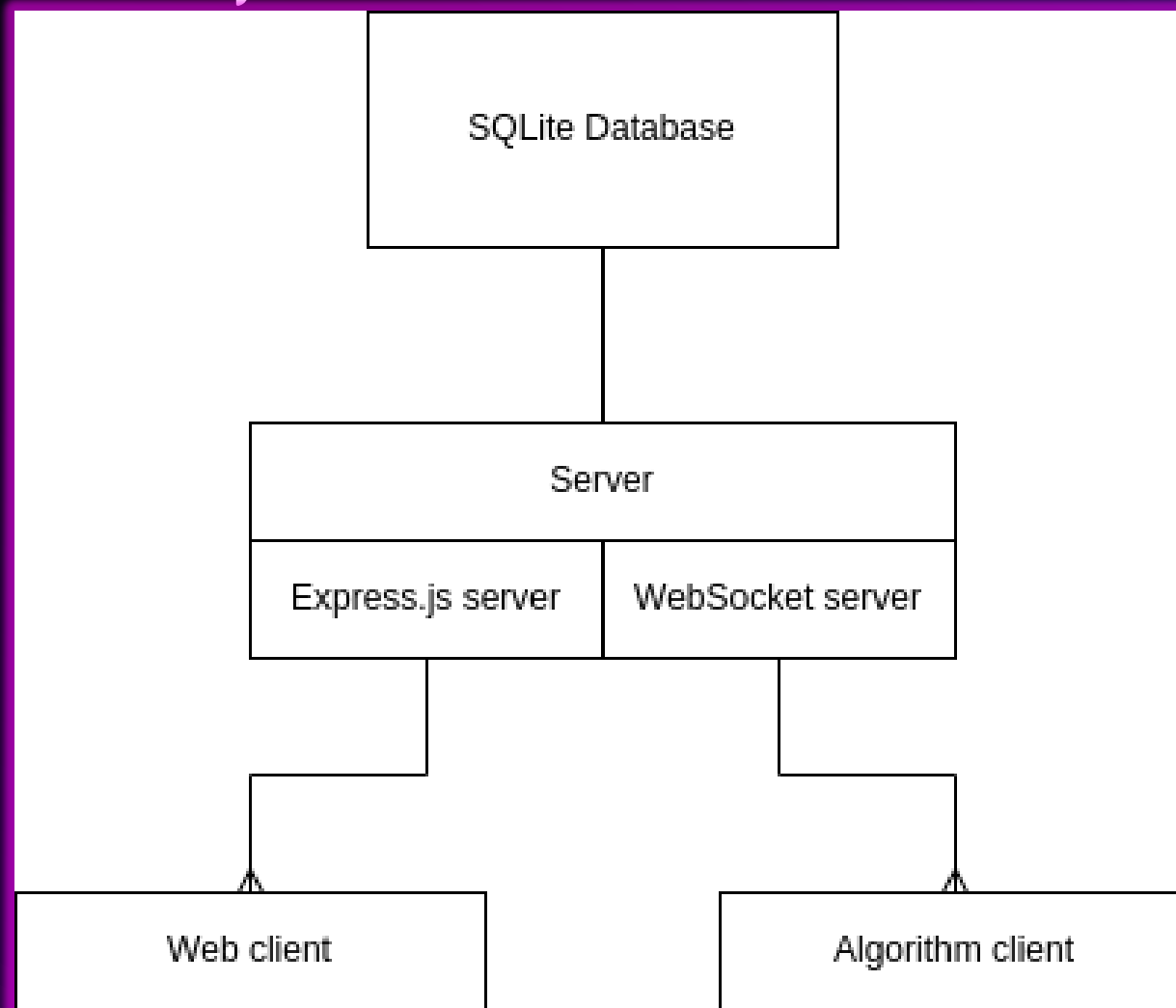
As an **Algorithm Designer**, I want to **develop** and **benchmark** algorithms to **streamline** optimisation of 3D robot navigation

Toshiba Developers

As a **Toshiba Developer**, I want to **explore** methods of storing goods in 3D space to **improve** efficiency of clients' business functions

Demonstration

Project Architecture



Tech Stack

Web Client

- Written in HTML, CSS and JavaScript
- Uses ThreeJS and WebGL to render boxes

Algorithm Client

- Python package
- Distributed as a wheel or source distribution

Server

- Written in TypeScript
- Run using NodeJS
- Communicates with client using WebSocket

Database

- Uses SQLite, stored locally

Decisions Behind Improvements

Old UI

Playground Replay Instructions Leaderboard Scenarios



No algorithms are currently running. See the instructions page to run your own.

Leaderboard

Main

This shows different sets of benchmarking across different scenarios.

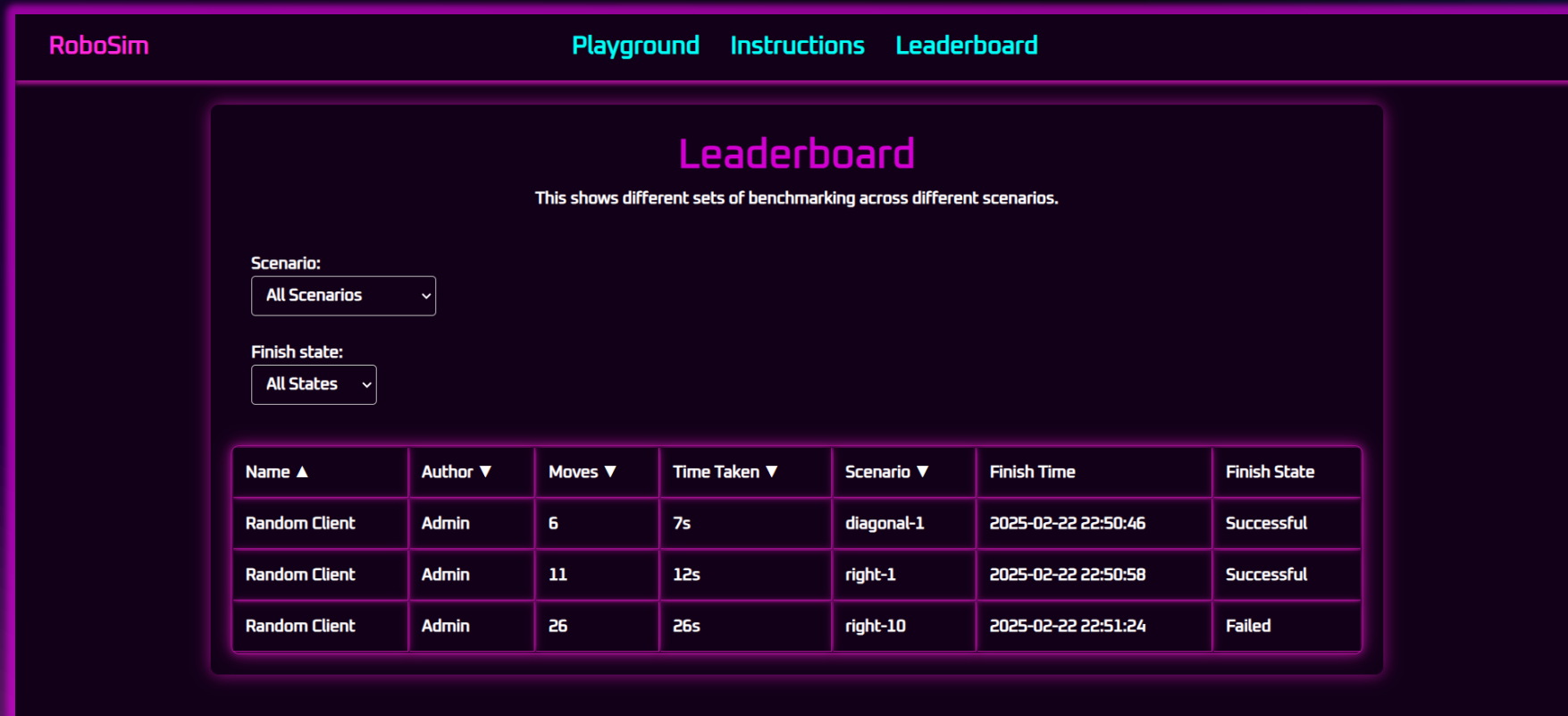
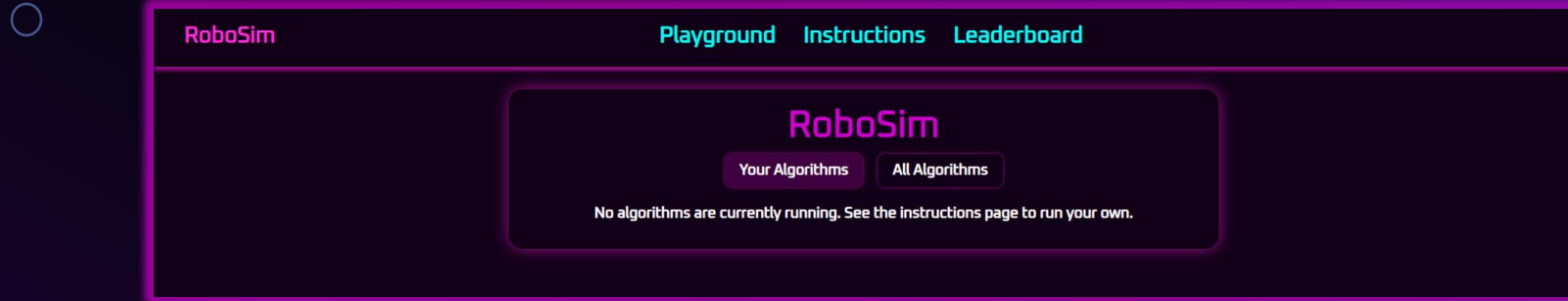
Scenario: DEMO_SCENARIO ▾

Sort by: Moves  ▾

SELECT

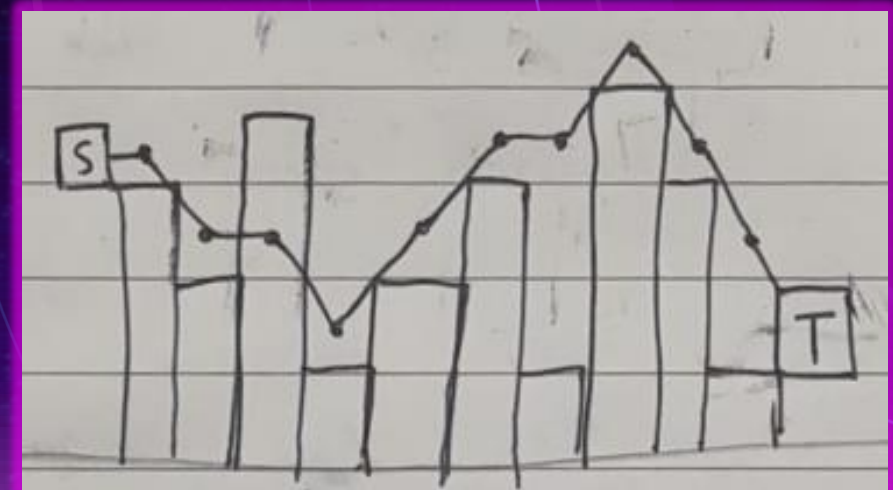
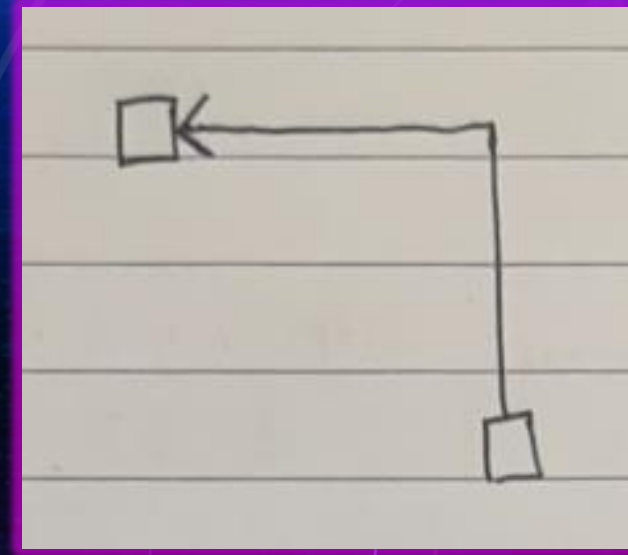
Name	Author	Datetime	Time Taken 	Moves 
Demo AS	Demo	2024-05-14 12:38:26	5	11
Kirans AS	Kiran	2024-10-09 13:07:00	5	11
Kirans AS	Kiran	2024-10-09 13:08:02	5	11

New UI



Navigation Algorithm

- Finds **top-down** path from source to destination
- Moves other boxes with **recursive** functions to make top-down path **accessible**



Leaderboard

- More intuitive leaderboard
- Order rows via columns
- Added pagination system

Before

Leaderboard 🏆

Main

This shows different sets of benchmarking across different scenarios.

Scenario: DEMO_SCENARIO Sort by: Moves 🦶 SELECT

Name	Author	Datetime	Time Taken 🦶	Moves 📱
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After

RoboSimPlaygroundInstructionsLeaderboard

Leaderboard

This shows different sets of benchmarking across different scenarios.

Scenario: All Scenarios

Finish state: All States

Name ▲	Author ▼	Moves ▼	Time Taken ▼	Scenario ▼	Finish Time	Finish State
Random Client	Admin	6	7s	diagonal-1	2025-02-22 22:50:46	Successful
Random Client	Admin	11	12s	right-1	2025-02-22 22:50:58	Successful
Random Client	Admin	26	26s	right-10	2025-02-22 22:51:24	Failed

Authentication

- Account-based system
- Users receive token on registration
- Account page to modify details

RoboSim Login

Username:

Password:

Login

Do not have an account? Register now!

User Profile

Username: admin



Name: Admin



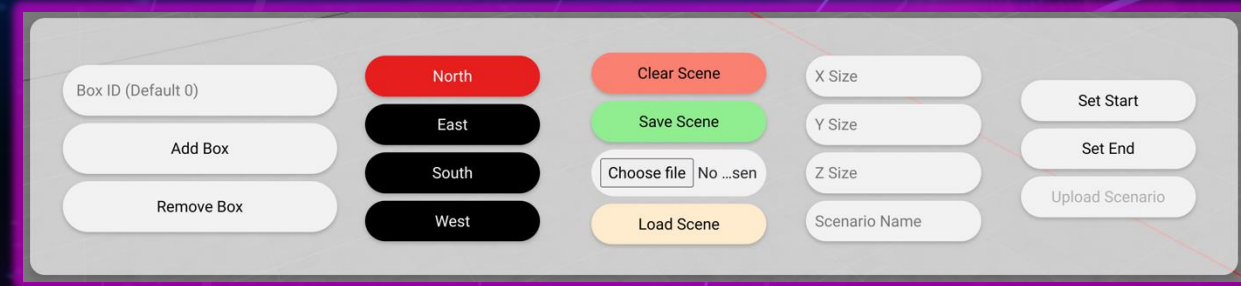
Change Password



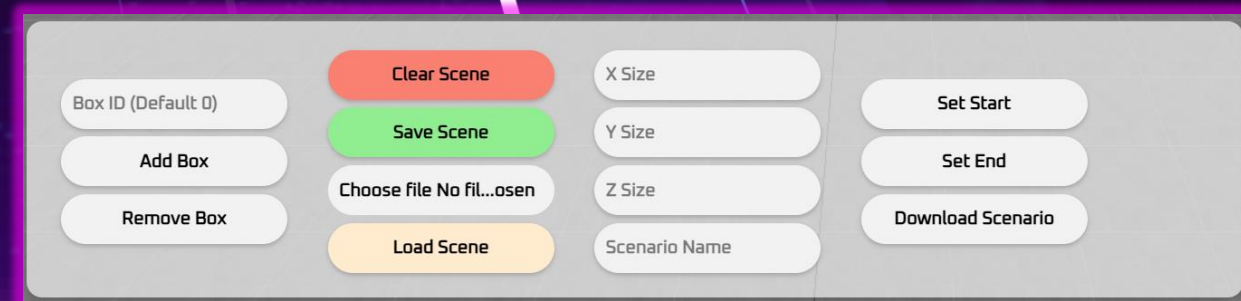
Playground (Part 1)

- Removed **redundant** cardinal buttons
- Box movement **solely** controlled via keypresses

Before

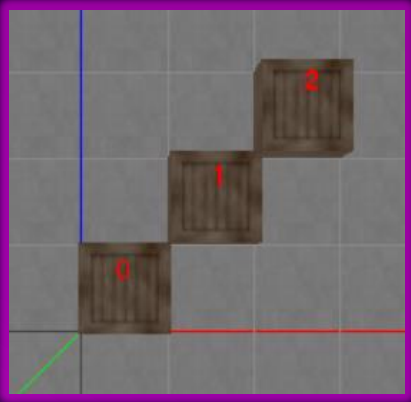


After



User Testing – Demo Scenes

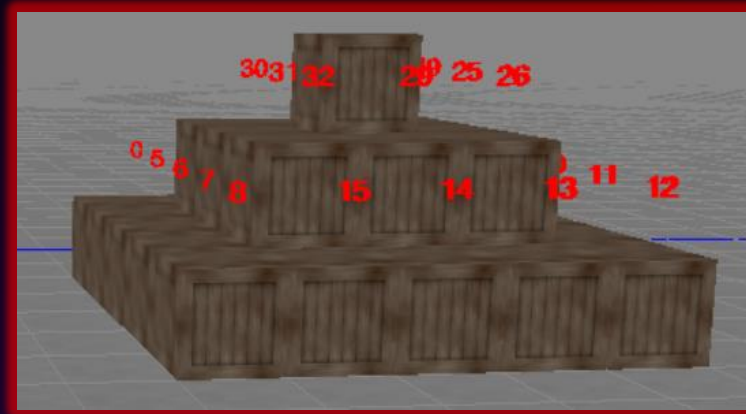
Scene 1



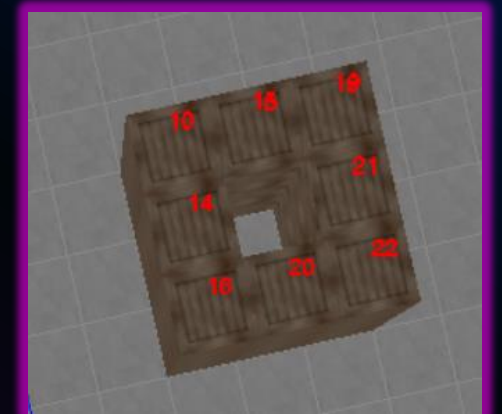
Scene 2



Scene 3



Scene 4



User Testing 1 – Quantitative

NASA-TLX



SUS



Mental Demand



Performance



Effort



Frustration



User Testing 1 – Qualitative

- Positives:
 - Fast rendering of boxes
 - Speedy movement of boxes
- Negatives:
 - Inconvenient to have to manually manage ID
 - Difficult to stack boxes
 - Box movement restrictions unexplained
 - Select Box feature not working

Playground (Part 2)

- Abstracted ID management from user
- Added **keypresses** for box creation & deletion
- Added **box restrictions** to Instructions

Before

The 'Before' state of the Playground interface shows a control panel with the following elements:

- Box ID (Default: 0)
- Add Box
- Remove Box
- Clear Scene (red button)
- Save Scene (green button)
- Choose file No fil...osen
- Load Scene (yellow button)
- X Size
- Y Size
- Z Size
- Scenario Name
- Set Start
- Set End
- Download Scenario

After

The 'After' state of the Playground interface shows a control panel with the following elements:

- Selected Box: None
- Add Box
- Remove Box
- Clear Scene (red button)
- Save Scene (green button)
- Choose file No fil...osen
- Load Scene (yellow button)
- X Size
- Y Size
- Z Size
- Scenario Name
- Set Start
- Set End
- Download Scenario
- Reset State

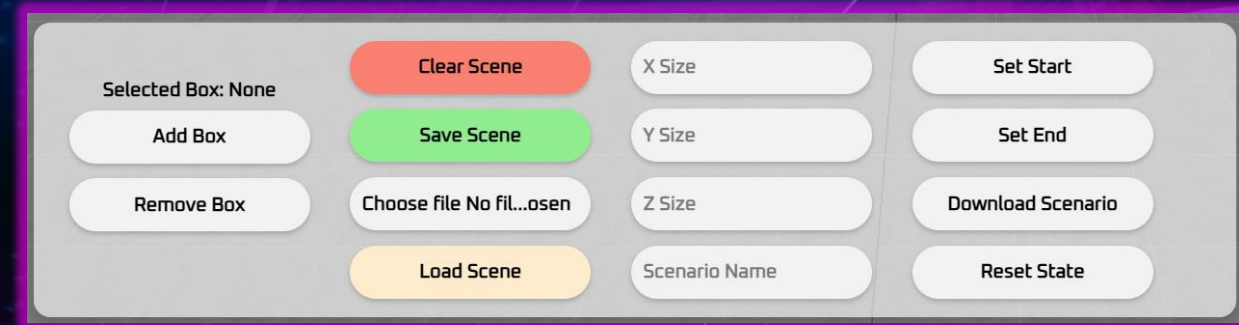
User Testing 2 – Qualitative

- Positives:
 - Automatic management of IDs
 - Engagement with all scenes
 - Speedy movement of boxes
- Negatives:
 - Difficult to stack boxes
 - Complex structures time-consuming to build
- Unanimous support for Forklift Mode (enables users to move stacks of boxes)

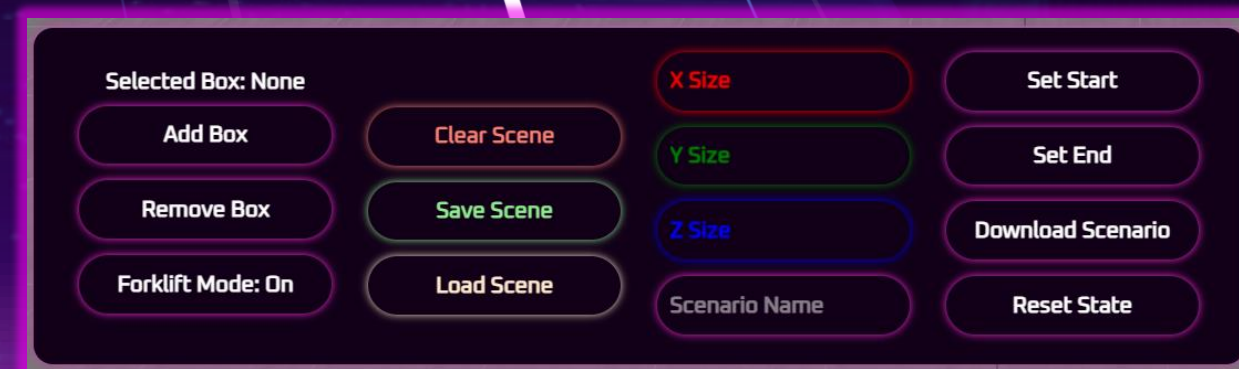
Playground (Part 3)

- Added **Forklift Mode** as a **toggle** feature (best compromise)
- **Styled** controls menu to with new colour scheme
- **Simplified** Load Scene functionality

Before

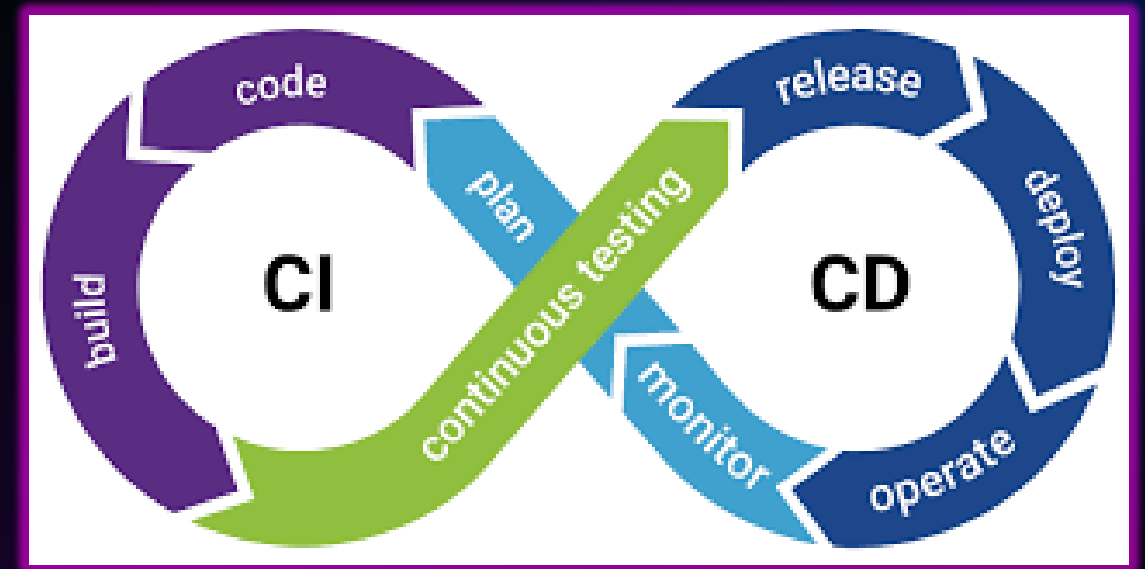


After



Testing & Deployment

- Improved CI
- Implemented CD
- Doubled number of tests on the server
- Increased code coverage to 87%



Client Feedback

- Positives:
 - Liked brand new UI
 - Appreciated navigation algorithm
 - Appreciated more tangible workload shown during meetings
- Improvements:
 - New modality of robotic box (optional)
 - Add dynamic but mellow element to webpages

Project Management (Part 1)

- Removed **redundant** Monday meetings
- Utilised more **channels** of communication
- Full **metadata** for issues & PRs
- **In-depth** code reviews

❑  **feat: User login system** ✓ **new feature** 1  54
#189 by DanielZhang05 was merged last month • Approved  Beta Release

❑  **feat: skip scenarios and use custom scenarios** ✓ **new feature** 2  30
#171 by si23870 was merged last month • Approved  Beta Release

❑  **feat: automate scenario generation** ✓ **new feature** **testing** 1  29
#243 by davidw1281 was merged 2 weeks ago • Approved  Beta Release

Todo 14

This item hasn't been started

2024-RoboSim #247

New main page

design new feature Priority: High

Size: Large Final Release #248

2024-RoboSim #151

Design a high-performance data structure for the warehouse.

new feature Priority: Medium

Size: Medium Final Release

2024-RoboSim #196

An idea about checking if the number of boxes are valid

In Progress 7

This is actively being worked on

2024-RoboSim #253

Refactor CSS files to use variables

refactor Priority: Medium Size: Medium

Final Release #306

2024-RoboSim #225

Add graph generation to leaderboard

new feature Priority: Medium

Size: Medium Final Release #281

2024-RoboSim #267

Add pagination to the dashboard and leaderboard

Review and test 1

Code/work to be looked over and tested

2024-RoboSim #289

Add pagination to the dashboard

enhancement Priority: Low Size: Small

Final Release

Done 169

This has been completed

2024-RoboSim #254

Add Forklift feature to Playground

new feature Priority: High Size: Medium

Final Release #293

2024-RoboSim #266

Account page

new feature Priority: Medium

Size: Medium Final Release #287

2024-RoboSim #153

Implement step 2 of the basic algorithm

new feature Priority: Medium Size: Large

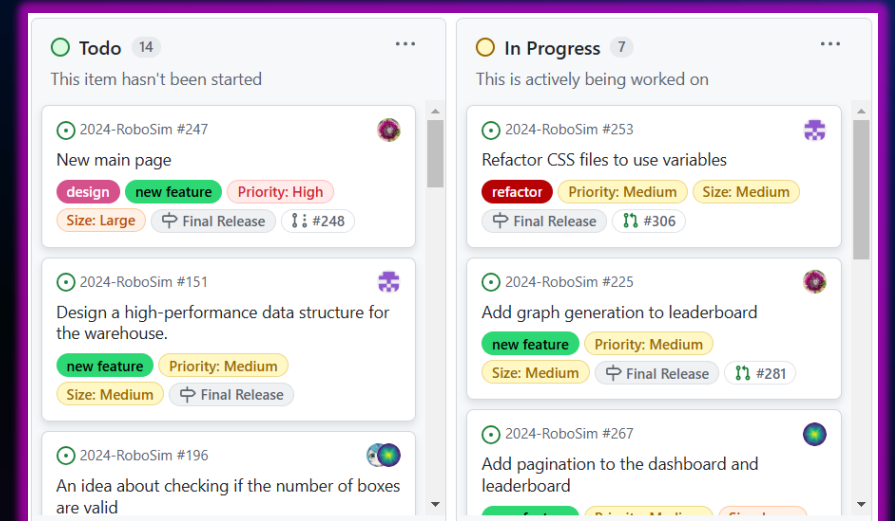
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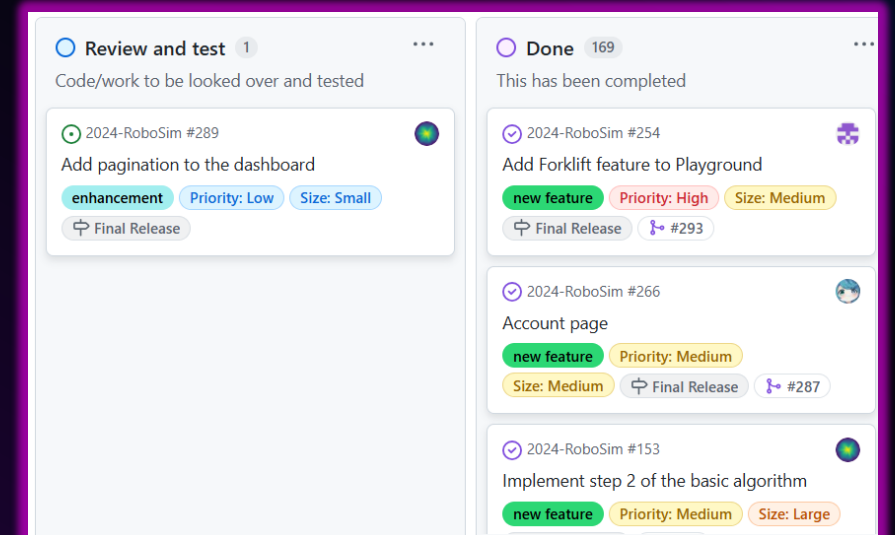
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The screenshot shows a Kanban board with two columns: 'Todo' (14 items) and 'In Progress' (7 items). The 'Todo' column contains three items: '2024-RoboSim #247' (New main page, design, new feature, Priority: High, Size: Large, Final Release, #248), '2024-RoboSim #151' (Design a high-performance data structure for the warehouse, new feature, Priority: Medium, Size: Medium, Final Release), and '2024-RoboSim #196' (An idea about checking if the number of boxes are valid). The 'In Progress' column contains three items: '2024-RoboSim #253' (Refactor CSS files to use variables, refactor, Priority: Medium, Size: Medium, Final Release, #306), '2024-RoboSim #225' (Add graph generation to leaderboard, new feature, Priority: Medium, Size: Medium, Final Release, #281), and '2024-RoboSim #267' (Add pagination to the dashboard and leaderboard).



The screenshot shows a Kanban board with two columns: 'Review and test' (1 item) and 'Done' (169 items). The 'Review and test' column contains one item: '2024-RoboSim #289' (Add pagination to the dashboard, enhancement, Priority: Low, Size: Small, Final Release). The 'Done' column contains three items: '2024-RoboSim #254' (Add Forklift feature to Playground, new feature, Priority: High, Size: Medium, Final Release, #293), '2024-RoboSim #266' (Account page, new feature, Priority: Medium, Size: Medium, Final Release, #287), and '2024-RoboSim #153' (Implement step 2 of the basic algorithm, new feature, Priority: Medium, Size: Large).

Project Management (Part 2)

- Time-bound tasks & use of Gantt chart
- Delegation of tasks from more experienced members
- AGILE methodologies for an iterative design process
- NO MORE SQUASHING COMMITS!!!!
- And one more addition...

Project Process Log

- The journey, in full
- Explains project-based decisions to future iterations
- Encourages proper documentation practices among group
- Easy to linearise for client at end of project
- Allows intangible workload to be recognised

Problems & Solutions

Problems

Losing a team member

More group disagreements

Solutions

Increased efficiency within team to mitigate impact and maintain work rate

Considered multiple angles to come to a holistic solution that maximised benefit

Plans for Final Release

- Finalise **dynamic** landing webpage
- **Update** Playground texture to support new theme
- A **new** modality of robotic box(?)



Thank You